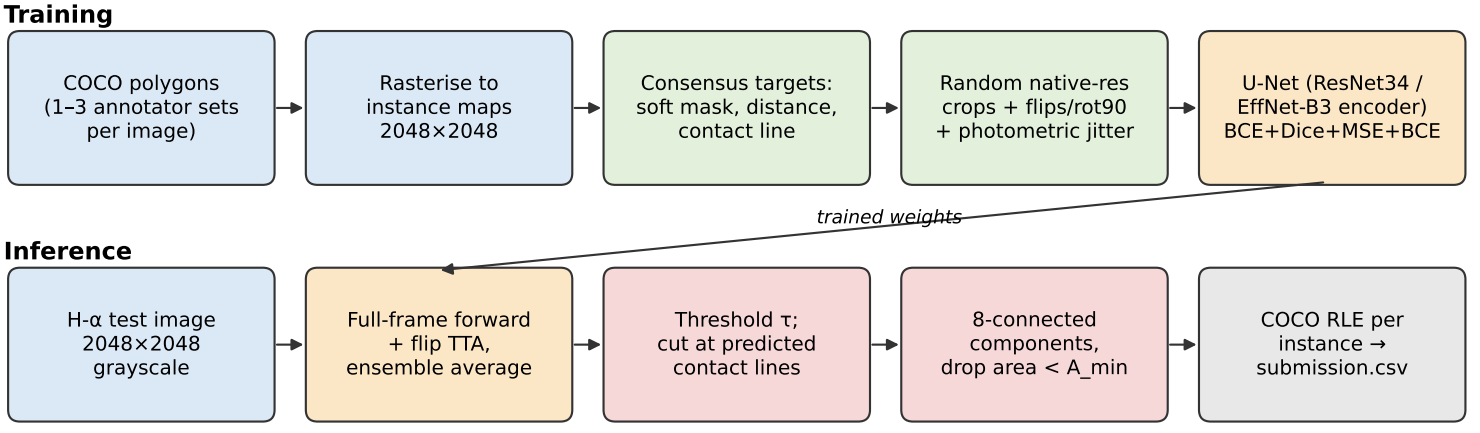
Training
COCO polygons (1–3 annotator sets per image)
Rasterise to instance maps 2048×2048
Consensus targets: soft mask, distance, contact line
Random native-res crops + flips/rot90 + photometric jitter
U-Net (ResNet34 / EffNet-B3 encoder) BCE+Dice+MSE+BCE
trained weights
Inference
H-α test image 2048×2048 grayscale
Full-frame forward + flip TTA, ensemble average
Threshold τ; cut at predicted contact lines
8-connected components, drop area < A_min
COCO RLE per instance → submission.csv